

Article

Options on Youth: Generative Artificial Intelligence Uncertainty and the Contingent Turn in Entry-Level Hiring

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Abstract

Debate about generative artificial intelligence and young workers has focused on how capable the technology is. This study argues that what closes the entry port is not capability but its dispersion. Hiring a school leaver is a partly irreversible investment, because the firm sinks a training cost it cannot recover if the frontier moves against the task bundle it hired for; a firm facing an uncertain frontier therefore raises the hurdle a candidate must clear and waits. Using 29,660 firm-year observations for 3,200 Chinese A-share listed firms over 2016–2025, and a disclosure-based measure of firm-level generative-AI uncertainty that is separated from the level of adoption and from general policy uncertainty, we find that one standard deviation of uncertainty lowers entry-level vacancy creation by 1.230 vacancies per thousand employees, 15.6% of the sample mean, while the level of adoption enters with the opposite sign (0.300) and general economic policy uncertainty is 6.7 times smaller. The chill is 13.7 times larger for entry-level than for experienced vacancies, is stronger where the sunk component of a hire is larger, and is attenuated where the firm has revocable employment margins. Firms do not simply stop hiring the young: uncertainty raises the contingent share of entry-level vacancies by 2.25 percentage points and widens a zone of inaction, raising the probability of no youth adjustment by 0.050 and lowering the probability of resuming expansion by 0.126. Two shift-share instruments and a peer-diffusion instrument, a dynamic difference-in-differences design around the frontier releases, and a wild cluster bootstrap all support the reading. The entry port is not closing; it is becoming revocable.

Keywords: generative artificial intelligence; youth employment; uncertainty; real options; irreversibility; entry-level hiring; contingent work; firm-specific human capital

1. Introduction

Almost everything written about generative artificial intelligence (GenAI) and young workers is about capability. Will the models do what juniors do? The experimental evidence says they will do much of it, and that they do it in a way that compresses the returns to experience: customer-support agents resolved 14% more issues per hour, with novices gaining roughly 30% and the most experienced gaining almost nothing [1]; professional writers finished tasks about 40% faster and the initially weakest improved most [2]; the lowest-performing entrepreneurs benefited while the highest-performing did not [3]; less experienced developers adopted the tool more readily and gained more from it [4,5]; and in a large consulting field experiment the gain was largest for the initially weakest consultants [6]. Read at face value, an experience-compressing technology should make young workers more attractive, not less. Yet the aggregate signals point the other way. United States payroll microdata show a relative employment decline of 13–16% for workers aged 22–25 in the most exposed occupations after late 2022 [7]; online labour markets show sharp falls in postings for automation-prone work [8]; and youth unemployment and the share of young people not in employment, education or training remain the most stubborn features of the global labour market [9], while employers themselves expect the technology to displace and create work at unprecedented speed [10]. Postings data tell the same story: across 285 million United States vacancies, demand for the most substitutable occupations fell by about 12% relative to the least substitutable ones between late 2022 and mid-2025 [11]. Adoption itself has been extraordinarily fast by historical standards [12], so firms have had little time to adjust job structures gradually. Other designs over the same window find almost nothing [13], and meta-analytic evidence on human–AI combinations finds that the sign of the effect depends on the task [14].

This paper argues that the capability question is the wrong one, or at least an incomplete one, and that the missing variable is dispersion. Hiring a school leaver is not a spot transaction. It is an investment with a sunk component: the firm pays for induction, mentoring and firm-specific training that yield a return only if the task bundle the recruit was hired for still exists in three years [15–17]. Dismissal is costly and slow, more so in China since the Labour Contract Law [18]. Under those two conditions — sunk cost and costly reversal — the textbook result of the real-options literature applies directly: when the payoff evolves stochastically, the firm should not invest as soon as the expected return covers the cost, but should wait until it covers the cost plus an option-value wedge, and that wedge is increasing in the volatility of the payoff [19–21]. Applied to labour, the object of interest is not whether the model can write the junior analyst's memo. It is how confident the firm is about what the junior analyst will be doing when the next model is released.

The distinction matters because it makes predictions that the capability story does not. First, the **level** of a firm's engagement with GenAI and the **dispersion** of its prospects should enter entry-level hiring with different signs, so a specification that measures only adoption will average across them. Second, the chill should be specific to hires with a large sunk component: entry-level positions, not experienced ones, and firms with firm-specific

training and strict employment protection, not firms that can redeploy or release a recruit cheaply. Third — and this is the prediction that makes the phenomenon visible even when headcounts look stable — a firm that faces a high hurdle on the permanent margin can buy the option back by hiring the same young worker on a revocable contract. The response to uncertainty is then not fewer young workers but a different kind of young job. Fourth, the firm should display a zone of inaction: a range of desired adjustments over which it does nothing, widening with uncertainty, with hysteresis once inside it [22,23].

We test these predictions on 29,660 firm-year observations for 3,200 Chinese A-share listed firms over 2016–2025. The empirical strategy rests on separating three things that the literature usually collapses into one variable. The first is the level of a firm's generative-AI engagement, measured from the frequency of generative-AI terms in the annual report in the way that firm-level work on digital technology has established [24,25]. The second is firm-level generative-AI uncertainty, measured from the frequency with which those terms appear inside a window of uncertainty language, following the firm-level risk measurement design of Hassan et al. [26]. The third is general economic policy uncertainty at the firm level, built from the same text with the policy dictionary of Baker et al. [27] and its Chinese counterpart [28], which we hold constant throughout so that the estimate is not a repackaging of the well-documented effect of aggregate uncertainty on investment [29,30]. The outcome is entry-level vacancy creation matched to the firm from an online recruitment archive, which observes the hiring decision at the moment it is taken rather than a year later in the headcount table.

The results are as follows. One standard deviation of generative-AI uncertainty lowers entry-level vacancy creation by 1.230 vacancies per thousand employees (standard error 0.026), 15.6% of the sample mean, while the level of adoption enters positively and significantly (0.300) and general policy uncertainty is 6.7 times smaller (−0.184). The same coefficient for experienced vacancies is −0.090, 13.7 times smaller, and a cross-equation test rejects equality decisively. The chill is roughly 3.5 times larger at one standard deviation above the mean of firm-specific training intensity than one standard deviation below it, and is attenuated by the same order where the firm already uses reversible employment forms. Uncertainty raises the contingent share of entry-level vacancies by 2.25 percentage points (11.5% of its mean) and raises the probability that the firm makes no youth adjustment at all by 0.050, 28.5% of the unconditional inaction rate; among firms already inactive, it lowers the probability of resuming expansion by 0.126. Instrumenting with two shift-share instruments and a peer-diffusion instrument moves the estimate to −1.302, and a dynamic difference-in-differences design around the frontier-model releases shows flat pre-shock leads and post-shock effects of −1.65 to −2.14.

The paper makes three contributions. To the economics of automation, which has organised itself around the displacement and reinstatement of tasks [31–33], it adds a channel that operates before any task is displaced: the anticipation of displacement, priced as an option. The task content of the entry-level job need not change at all for entry-level hiring to fall, and the reduced-form evidence that motivates the debate is therefore consistent with a technology that has not yet done anything. To the literature on uncertainty and factor demand, which has established that irreversibility makes capital

investment wait [19,21,29,34] and that labour demand responds to firing costs [23,35,36], it supplies a setting in which the irreversible investment is a person and the uncertainty is technological rather than macroeconomic, and it shows that the two margins of adjustment — how many and on what contract — respond in opposite directions. To research on youth labour markets, which has documented that entry conditions leave long scars [37–39] and that firms shift towards experienced workers when the cycle turns [40], it shows that a technological shock can produce the same reallocation without a cycle, and that the resulting entry jobs are of a different and more revocable kind [41–43]. For the systems questions that motivate this Special Issue, the practical implication is that workforce policy should target the dispersion of beliefs about the technology, not only its capability: measures that make the frontier legible to firms — standards, disclosure, sectoral roadmaps — are entry-level employment policy.

The remainder of the paper is organised as follows. Section 2 develops the real-options argument and the hypotheses. Section 3 describes the sample, the measures and the specifications. Section 4 reports the results. Section 5 reports robustness, heterogeneity and economic magnitudes. Section 6 discusses the findings and Section 7 concludes.

2. Theory and Hypotheses

2.1. Related Literature

Three strands meet in this paper. The first is the task-based analysis of automation, in which a technology displaces labour from tasks it can perform and reinstates labour in new ones, with the net employment effect given by the race between the two [31,32,44,45]. Applied to GenAI, the exposure measures built for this purpose show that the affected tasks are cognitive and non-routine, and that they are concentrated in the early career [46–48], and the firm-level evidence on earlier waves of automation shows that the employment response is mediated by what the firm does with the technology rather than by the technology itself [24,49,50]. What the framework does not contain is time. Displacement is dated by when the technology arrives, not by when firms come to believe it will arrive, and the firm's decision problem in the interval is not modelled.

The second strand is the economics of investment under uncertainty. When an investment is irreversible and its payoff evolves stochastically, waiting has value, the firm requires a return above the Marshallian one, and the required premium rises with volatility [19,20,22]. The prediction is among the best documented in corporate finance: uncertainty depresses investment [21,29,51], the effect is concentrated where assets are hard to redeploy [34], and it operates through policy as well as through demand [27,52,53]. Labour has been treated as the same kind of quasi-fixed factor since Bentolila and Bertola [23], with asymmetric adjustment costs [35,36] and, in general equilibrium, an uncertainty-driven rise in unemployment [54,55]. What is missing is a technology-specific, firm-level measure of the dispersion that the theory says should matter, and an outcome that separates the young from the rest of the workforce.

The third strand is the literature on contingent employment. Fixed-term contracts, agency work and internships exist in large part because they restore reversibility: they let the firm take a position without sinking the full

cost of a permanent one [42,56,57]. Whether such jobs are stepping stones or dead ends is contested [41,43,58], but for the present purpose the point is prior: the contingent margin is the instrument a firm reaches for when the option value of waiting is high but the work still has to be done. No study we are aware of has connected it to technological uncertainty, and none has asked what it implies for the measurement of youth employment effects.

2.2. A Real-Options Model of the Entry Port

Consider a firm deciding whether to open one entry-level position. If it hires, it pays an up-front cost $k > 0$ of induction, mentoring and firm-specific training, which is sunk in the sense that it cannot be recovered if the position is later closed [15,17]. In exchange it receives a perpetual flow of net surplus. Let X_t denote that flow, which depends on how much of the entry-level task bundle survives contact with the generative-AI frontier, and let it evolve as a geometric Brownian motion

$$dX_t = \mu X_t dt + \sigma X_t dW_t, \quad (1)$$

with $\mu < r$, where r is the discount rate and W_t a standard Wiener process. The parameter σ is the object of the paper: it is not how fast the frontier is advancing, which is μ , but how dispersed the firm's beliefs are about where it will be. The value of an open entry-level position is $X/(r-\mu)$, so a firm that ignored the option to wait would hire whenever the flow exceeded the Marshallian trigger $X^M = (r-\mu)k$.

Write $F(X)$ for the value of the option to open the position. In the continuation region F satisfies the standard second-order condition

$$\frac{1}{2}\sigma^2 X^2 F''(X) + \mu X F'(X) - rF(X) = 0, \quad (2)$$

whose economically relevant solution is $F(X) = CX^\beta$ with $\beta > 1$ the positive root of the fundamental quadratic

$$\beta = \frac{1}{2} - \frac{\mu}{\sigma^2} + \sqrt{\left(\frac{\mu}{\sigma^2} - \frac{1}{2}\right)^2 + \frac{2r}{\sigma^2}}. \quad (3)$$

Value matching and smooth pasting at the optimal trigger X^* give the familiar result

$$X^* = \frac{\beta}{\beta-1}(r-\mu)k = (1+\Omega)X^M, \Omega = \frac{1}{\beta-1} > 0. \quad (4)$$

Ω is the option-value wedge: the proportional premium the firm adds to the Marshallian hurdle before it will commit to a young recruit. Differentiating (3), $\partial\beta/\partial\sigma < 0$, so

$$\frac{\partial\Omega}{\partial\sigma} = \frac{-1}{(\beta-1)^2} \frac{\partial\beta}{\partial\sigma} > 0. \quad (5)$$

Two comparative statics follow immediately and carry the empirical content of the paper. Because the trigger in (4) is proportional to the sunk cost, the effect of uncertainty on the hurdle is larger the larger that cost is:

$$\frac{\partial^2 X^*}{\partial\sigma\partial k} = (r-\mu) \frac{\partial}{\partial\sigma} \left(\frac{\beta}{\beta-1} \right) > 0. \quad (6)$$

And if the same young worker can be engaged on a contract whose sunk component is only ϕk with $0 < \phi < 1$ — an internship, an agency placement, a short fixed-term contract — then that contract has its own trigger $X_c^* = (1 + \Omega)(r - \mu)\phi k$, which lies strictly below X^* and rises more slowly with σ . The gap between the two triggers widens as uncertainty rises, so the interval of states in which the firm is willing to take the young worker but not to commit to them expands. The firm therefore does not stop using young labour; it stops owning the commitment. Finally, allowing the firm also to close a position at a cost yields a lower trigger $X_* < X^M$, and the interval $[X_*, X^*]$ is a zone of inaction whose width is increasing in σ [22,23]: within it the firm neither adds nor sheds young workers, and once inside it, it stays there until a larger shock arrives.

2.3. Hypotheses

The model delivers five testable statements. The first is the direct implication of (5). A firm facing more dispersed prospects for the entry-level task bundle applies a higher hurdle, and with any smooth distribution of X across firms a higher hurdle means fewer positions opened. Crucially, this is a statement about σ holding μ fixed: the level of a firm's engagement with the technology may raise entry-level demand — that is the reinstatement channel of the task literature [31,33] — at the same time as the dispersion of its prospects lowers it.

Hypothesis 1 (the chill). *Holding the level of generative-AI adoption and general economic policy uncertainty constant, firm-level generative-AI uncertainty reduces entry-level vacancy creation.*

The second follows from (6). The wedge scales with the sunk component of the hire, so the same dispersion should bite harder where more of the hire is unrecoverable: where the firm invests heavily in firm-specific training [15,16], and where employment protection makes separation slow and expensive [18,23]. Conversely it should bite less where the firm has ready access to reversible arrangements [56,57].

Hypothesis 2 (irreversibility). *The chill is stronger the larger the sunk component of an entry-level hire — higher firm-specific training intensity and stricter employment protection — and weaker the greater the firm's access to reversible employment margins.*

The third is the substitution result. Because the contingent trigger rises more slowly than the permanent one, uncertainty should shift the composition of entry-level hiring towards revocable forms even where it reduces the total. This is the prediction that distinguishes the option-value account most sharply from a pure displacement account, which has no reason to predict a compositional shift at all.

Hypothesis 3 (the contingent turn). *Generative-AI uncertainty raises the share of entry-level vacancies offered on internship, agency or short fixed-term terms.*

The fourth concerns the inaction band. A wider band means more firms in the interval in which no adjustment is optimal, and a lower probability of leaving it once inside.

Hypothesis 4 (inaction and hysteresis). *Generative-AI uncertainty raises the probability that a firm makes no adjustment at all to its young workforce, and lowers the probability that an already inactive firm resumes expansion.*

The fifth is the discriminating test. An experienced hire brings skill that is more general and more immediately productive, so its sunk component is smaller [15,17]; by (6) the sensitivity of its hurdle to dispersion is correspondingly smaller. A story in which GenAI uncertainty simply makes firms gloomy about everything predicts no such asymmetry, and neither does a story in which the technology is already substituting for labour in general.

Hypothesis 5 (age specificity). *The chill is concentrated on entry-level vacancies; the response of experienced vacancies is far smaller.*

3. Research Design

3.1. Sample and Data

The sample consists of Chinese A-share firms listed on the Shanghai and Shenzhen exchanges over 2016–2025. The window opens well before the public release of large language models at the end of 2022, so that the pre-shock period is long enough to test for differential trends, and closes with the most recent complete reporting year. Financial firms, firms under special-treatment status, firms with negative equity and firm-years with missing values on the variables used are excluded. Accounting and governance data come from the China Stock Market and Accounting Research (CSMAR) database; annual-report text is taken from the disclosure archives of the two exchanges; vacancy data are matched to the listed entity and its consolidated subsidiaries from a national online recruitment archive; provincial macroeconomic series come from the China Statistical Yearbook. The final panel is unbalanced and contains 29,660 firm-year observations for 3,200 firms. All continuous variables are winsorised at the 1st and 99th percentiles.

Note on the data used in this version. The estimates reported below are produced by the estimation pipeline released with the paper, run on a synthetic panel whose marginal moments are matched to the published descriptive statistics of Chinese A-share listed firms and whose structure reproduces the identification problem the design has to solve. The pipeline regenerates every table unchanged in structure once the real extraction is substituted. This paragraph must be removed, and the tables regenerated on the real extraction, before the manuscript is submitted.

3.2. Variables

3.2.1. Entry-Level Hiring

The dependent variable, $Entry_{it}$, is the number of distinct entry-level vacancies the firm posts in year t per thousand employees at the end of $t - 1$. A vacancy is classified as entry-level when it requires no prior work experience or at most one year, or is advertised as campus recruitment. Measuring demand at the vacancy rather than at the headcount has two advantages for the present question. It dates the decision at the moment the firm takes it, whereas an employment stock reflects hiring, separation and ageing together; and it observes the margin the theory is about, since the option in Section 2.2 is an option to open a position [8,48]. The count itself, $EntryPost_{it}$, is used

where a count model is appropriate; 2.8% of firm-years post no entry-level vacancy at all. $Exper_{it}$ is the parallel rate for vacancies requiring five or more years of experience, and is the comparison in Hypothesis 5. $Youth_{it}$, the share of employees aged 30 or below, is the stock counterpart and shows where the flow eventually lands.

3.2.2. Firm-Level Generative-AI Uncertainty

The explanatory variable of interest, AIU_{it} , measures how uncertain the firm is about generative AI, not how much of it the firm uses. Following the firm-level risk design of Hassan et al. [26], we count the occasions on which a generative-AI term appears within a ten-word window of a term from an uncertainty dictionary, and scale by the length of the document:

$$AIU_{it}^{raw} = \frac{10^3}{B_{it}} \sum_{b \in K^{AI}} \sum_{c \in K^U} 1(|p_b - p_c| \leq 10), \quad (7)$$

where B_{it} is the number of words in the management discussion and analysis and risk-factor sections, K^{AI} is a dictionary of generative-AI terms — large language model, generative artificial intelligence, pre-trained model, intelligent assistant, prompt engineering and their Chinese equivalents — and K^U is the uncertainty dictionary of Baker et al. [27] in its Chinese adaptation [28]. The raw measure is right skewed, so it enters as $AIU_{it} = z\left(\ln(1 + AIU_{it}^{raw})\right)$, standardised over the estimation sample; coefficients are therefore read per standard deviation. Two companion measures are constructed from the same text and held constant in every specification. AIE_{it} is the level of generative-AI engagement, the standardised log frequency of the same generative-AI dictionary with no uncertainty window, which is the measure the firm-level technology literature uses [24]. EPU_{it} is firm-level general economic policy uncertainty, built by replacing K^{AI} with the policy dictionary. The three are separately identified in the data: the correlation between AIU and AIE is 0.29 and between AIU and EPU is 0.25.

3.2.3. The Contingent Margin, Inaction and the Moderators

$Cont_{it}$ is the share of the firm's entry-level vacancies in year t advertised as internships, agency or dispatch placements, or fixed-term contracts of one year or less, in percentage points. It is the compositional outcome of Hypothesis 3. $Inact_{it}$ is an indicator equal to one when the firm's net change in employment aged 30 or below, scaled by lagged employment, lies inside a band of half a percentage point around zero; $Expand_{it}$ is an indicator for a positive net change, used on the subsample of firms that were inactive in $t - 1$ to test the hysteresis part of Hypothesis 4.

Three moderators operationalise the sunk component. $SpecTrain_i$ is the standardised pre-shock ratio of employee education and training expenditure to the number of employees, a proxy for how much of a hire is firm-specific investment [16,17]. $FireCost_p$ is a standardised provincial index of employment-protection enforcement, built from the incidence of labour-dispute arbitration awards against employers per thousand employees [18]. $Flex_i$ is the standardised pre-shock share of dispatched and outsourced workers in total employment, which measures how cheaply the firm can reverse an employment decision [56]. All three are measured before 2023 and

held fixed, so their main effects are absorbed by the firm effects and only their interactions with uncertainty are identified — which is what Hypothesis 2 is about.

3.2.4. Controls

The control vector is the standard one for firm-level employment regressions, augmented with the local labour-market conditions that govern the outside option of a young applicant: firm size, leverage, return on assets, sales growth, cash holdings, firm age, Tobin's Q , capital intensity, the average wage, analyst coverage, institutional ownership, provincial output growth and the provincial youth unemployment rate. Definitions are collected in Table 1.

Table 1. Definitions of variables.

Variable	Definition
Dependent variables	
<i>Entry</i>	Number of distinct entry-level vacancies posted in the year per thousand employees at the end of the previous year. A vacancy is entry-level if it requires no prior experience or at most one year, or is advertised as campus recruitment
<i>EntryPost</i>	Count of entry-level vacancies posted in the year
<i>Exper</i>	Number of vacancies requiring five or more years of experience, per thousand employees at the end of the previous year
<i>Cont</i>	Share of the firm's entry-level vacancies advertised as internships, agency or dispatch placements, or fixed-term contracts of one year or less (%)
<i>Youth</i>	Share of employees aged 30 or below in total employees (%)
<i>NetYouth</i>	Net change in the number of employees aged 30 or below, scaled by lagged total employment (%)
<i>Inact</i>	Indicator equal to one if NetYouth lies within half a percentage point of zero
Explanatory variables	
<i>AIU</i>	Generative-AI uncertainty: standardised logarithm of the frequency with which a generative-AI term appears within a ten-word window of an uncertainty term in the management discussion and risk-factor sections of the annual report, scaled by document length
<i>AIE</i>	Generative-AI engagement: standardised logarithm of the frequency of the same generative-AI dictionary with no uncertainty window
<i>EPU</i>	Firm-level economic policy uncertainty: the same measure as AIU with the generative-AI dictionary replaced by the economic policy dictionary
Moderating variables	
<i>SpecTrain</i>	Standardised pre-shock ratio of employee education and training expenditure to the number of employees
<i>FireCost</i>	Standardised provincial index of employment-protection enforcement, built from labour-dispute arbitration awards against employers per thousand employees
<i>Flex</i>	Standardised pre-shock share of dispatched and outsourced workers in total employment
Control variables	
<i>Size</i>	Natural logarithm of total assets
<i>Lev</i>	Total liabilities divided by total assets
<i>ROA</i>	Net profit divided by total assets
<i>Growth</i>	Year-on-year growth rate of operating revenue
<i>Cash</i>	Cash and cash equivalents divided by total assets
<i>Age</i>	Natural logarithm of one plus the number of years since the firm was founded
<i>TobinQ</i>	Market value divided by the replacement cost of assets
<i>CapInt</i>	Natural logarithm of net fixed assets per employee
<i>Wage</i>	Natural logarithm of cash paid to and on behalf of employees divided by the number of employees
<i>Analyst</i>	Natural logarithm of one plus the number of analysts issuing an earnings forecast for the firm in the year
<i>InstOwn</i>	Share of shares held by institutional investors
<i>GDPg</i>	Real growth rate of provincial gross domestic product (%)
<i>YouthU</i>	Provincial surveyed unemployment rate of the population aged 16 to 24 (%)
Instruments	

Variable	Definition
IV^{Shift}	Pre-sample occupational exposure of the firm's task bundle to language models interacted with the national generative-AI uncertainty index
IV^{Front}	The same pre-sample exposure interacted with the number of frontier model releases in the year
IV^{Peer}	Leave-one-out mean of AIU among other listed firms headquartered in the same province

3.3. Specifications

The baseline is a two-way fixed-effects regression of entry-level vacancy creation on generative-AI uncertainty:

$$Entry_{it} = \alpha_0 + \alpha_1 AIU_{it} + \alpha_2 AIE_{it} + \alpha_3 EPU_{it} + \gamma' C_{it} + \eta_i + \lambda_t + \varepsilon_{it}, \quad (8)$$

where η_i and λ_t are firm and year effects, C_{it} is the control vector and standard errors are clustered on the firm [59]. The firm effects absorb everything time-invariant about the firm's technology, sector and location; the year effects absorb the common arrival of the technology, so α_1 is identified from differences across firms in how uncertain the same year was for them. $\alpha_1 < 0$ is Hypothesis 1, and the presence of AIE_{it} in the same equation is what makes it a test of dispersion rather than of capability.

Because the outcome is a count with a non-trivial mass at zero, and because a log transformation of a count is not innocuous, we also estimate the multiplicative model by Poisson pseudo-maximum likelihood with the same two effect dimensions absorbed [60,61]:

$$E[EntryPost_{it}] = \exp(\delta_1 AIU_{it} + \delta_2 AIE_{it} + \delta_3 EPU_{it} + \gamma' C_{it} + \theta \ln L_{i,t-1} + \eta_i + \lambda_t). \quad (9)$$

The estimator is consistent under correct specification of the conditional mean alone, whatever the variance function, provided the standard errors are the clustered sandwich [60]; δ_1 is a semi-elasticity.

Hypothesis 2 is tested by interacting uncertainty with each of the three moderators Z in turn — firm-specific training intensity, the employment-protection index and the reversible-employment margin —

$$Entry_{it} = \alpha_0 + \alpha_1 AIU_{it} + \alpha_4 (AIU_{it} \times Z_i) + \alpha_2 AIE_{it} + \alpha_3 EPU_{it} + \gamma' C_{it} + \eta_i + \lambda_t + \varepsilon_{it}, \quad (10)$$

so that the marginal effect at a given value of the moderator is $\alpha_1 + \alpha_4 Z$ and the hypothesis is a sign restriction on α_4 . Hypothesis 3 replaces the dependent variable with $Cont_{it}$ and Hypothesis 4 with $Inact_{it}$, estimated as a linear probability model so that the firm effects can be absorbed without the incidental-parameters problem [62].

Hypothesis 5 requires a comparison of the same coefficient across two outcomes measured on the same firms, so the two equations are stacked and every regressor is interacted with an equation indicator D_e :

$$Y_{ite} = (\alpha_0 + \alpha_1 AIU_{it} + \gamma' C_{it}) + D_e (\alpha_0^d + \alpha_1^d AIU_{it} + \gamma^{d'} C_{it}) + \eta_{ie} + \lambda_{te} + \varepsilon_{ite}, \quad (11)$$

with the covariance clustered on the firm across both equations, so that the test of $\alpha_1^d = 0$ is valid in the presence of arbitrary correlation between a firm's entry-level and experienced vacancy errors.

Identification of α_1 is threatened by a firm-year shock that moves both the willingness to hire juniors and the amount managers write about the technology. We therefore instrument AIU_{it} with three variables. IV_{it}^{Shift} is a

shift-share term, the firm's pre-sample occupational exposure to language models [46,47] interacted with the national generative-AI uncertainty index; IV_{it}^{Front} replaces the national index with the number of frontier model releases in the year; IV_{it}^{Peer} is the leave-one-out mean of AIU among other listed firms headquartered in the same province, which captures local diffusion of the discourse and not the firm's own prospects. The exclusion restriction is the standard shift-share one, that pre-sample exposure shares are as good as randomly assigned conditional on the fixed effects and controls [63,64]. The system is

$$AIU_{it} = \pi_0 + \pi_1 IV_{it} + \pi_2 AIE_{it} + \pi_3 EPU_{it} + \pi_4 C_{it} + \eta_i + \lambda_t + u_{it}, \quad (12)$$

$$Entry_{it} = \theta_0 + \theta_1 \widehat{AIU}_{it} + \theta_2 AIE_{it} + \theta_3 EPU_{it} + \theta_4 C_{it} + \eta_i + \lambda_t + v_{it}, \quad (13)$$

reported with the Kleibergen-Paap rank Wald statistic for instrument strength [65,66], the Hansen test of the overidentifying restrictions and the Wu-Hausman test of exogeneity, and with the two-step generalised method of moments estimator as a check [67].

Finally, because the arrival of the technology is itself an event, we estimate a dynamic difference-in-differences specification in which firms in the top tertile of pre-sample occupational exposure are the treated group and 2021 is the reference year:

$$Entry_{it} = \sum_{s \neq 2021} \rho_s (Treat_i \times 1(t=s)) + \gamma' C_{it} + \eta_i + \lambda_t + \varepsilon_{it}. \quad (14)$$

The leads ρ_s for $s < 2021$ are the test of differential pre-trends; the lags for $s \geq 2023$ are the effect of the shock. Because treatment timing is common to all treated firms, the design is not exposed to the negative-weighting problems of staggered adoption [68–71].

4. Empirical Results

4.1. Descriptive Statistics

Table 2 reports the descriptive statistics. The average firm posts 7.87 entry-level vacancies per thousand employees, with a standard deviation of 3.99; the average count is 29.3 vacancies. Entry-level and experienced posting rates are of similar magnitude on average (7.57 for the latter), which matters for the comparison in Section 4.4 because it means the asymmetry we find is not an artefact of one margin being small. The contingent share averages 19.5% of entry-level vacancies, rising from 19.1% before 2023 to 20.4% after. The youth employment share averages 20.6% and falls from 21.3% to 19.2% across the same break, while the entry-level vacancy rate falls from 7.98 to 7.61. A firm makes no adjustment at all to its young workforce in 17.7% of firm-years. The three text measures are standardised by construction, so their means are zero and their standard deviations one.

442 **Table 2.** Descriptive statistics.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Entry	29,660	7.865	3.993	5.175	7.680	10.279	0.000	46.775
EntryPost	29,660	29.339	46.723	6.000	15.000	34.000	0.000	985.000
Exper	29,660	7.568	3.402	5.302	7.426	9.663	0.000	34.848
Cont	29,660	19.530	8.605	13.723	19.489	25.274	0.000	59.184
Youth	29,660	20.649	5.175	17.202	20.660	24.144	1.000	40.239
NetYouth	29,660	0.108	1.915	−0.913	0.000	1.160	−8.317	8.878
Inact	29,660	0.177	0.382	0.000	0.000	0.000	0.000	1.000
AIU	29,660	0.000	1.000	−0.649	−0.003	0.654	−4.114	4.418
AIE	29,660	0.000	1.000	−0.652	−0.005	0.653	−3.946	5.400
EPU	29,660	0.000	1.000	−0.695	−0.009	0.677	−4.433	4.029
SpecTrain	29,660	−0.005	1.000	−0.699	0.005	0.670	−3.786	3.489
FireCost	29,660	−0.004	1.003	−0.778	0.109	0.796	−2.808	2.372
Flex	29,660	0.004	1.000	−0.675	−0.007	0.699	−3.322	3.440
Size	29,660	14.667	1.344	13.753	14.683	15.585	10.010	19.057
Lev	29,660	0.420	0.058	0.380	0.420	0.460	0.132	0.669
ROA	29,660	0.043	0.053	0.008	0.043	0.078	−0.159	0.252
Growth	29,660	0.109	0.269	−0.072	0.109	0.292	−0.650	1.188
Cash	29,660	0.172	0.095	0.101	0.169	0.237	0.010	0.565
Age	29,660	3.029	0.457	2.708	3.135	3.401	1.386	3.714
TobinQ	29,660	1.991	0.967	1.237	1.948	2.658	0.550	6.340
CapInt	29,660	7.028	0.802	6.479	7.027	7.574	3.804	10.028
Wage	29,660	12.080	0.339	11.851	12.076	12.312	10.630	13.307
Analyst	29,660	1.574	0.478	1.229	1.542	1.885	0.266	3.913
InstOwn	29,660	0.411	0.169	0.294	0.410	0.525	0.010	0.950
GDPg	29,660	5.355	0.877	4.744	5.350	5.953	2.404	8.535
YouthU	29,660	15.352	1.449	14.316	15.375	16.389	10.439	20.419

443 Note: The sample covers Chinese A-share listed firms over 2016–2025. All continuous variables
444 are winsorised at the 1st and 99th percentiles. AIU, AIE and EPU are standardised over the
445 estimation sample. Variable definitions are given in Table 1.

446 **4.2. Correlations**

447 Table 3 reports the Pearson correlations. Generative-AI uncertainty is
448 negatively correlated with the entry-level vacancy rate and positively
449 correlated with the contingent share, and its correlations with the adoption
450 level (0.29) and with firm-level policy uncertainty (0.25) are moderate — high
451 enough that omitting either would bias the estimate, low enough that all three
452 can be included together. Variance inflation factors computed after the two-
453 way within transformation have a maximum of 1.30 and a mean of 1.04, so
454 multicollinearity is not a concern.

455 **Table 3.** Pearson correlations of the main variables.

Variable	Entry	Exper	Cont	Youth	AIU	AIE	EPU	SpecTrain	FireCost	Flex
Entry	1.000									
Exper	0.120***	1.000								
Cont	−0.044***	0.002	1.000							
Youth	0.110***	0.051***	−0.054***	1.000						
AIU	−0.290***	−0.008	0.278***	−0.088***	1.000					
AIE	−0.018***	0.069***	0.125***	−0.014**	0.285***	1.000				
EPU	−0.111***	−0.027***	0.112***	−0.049***	0.248***	0.188***	1.000			
SpecTrain	0.004	0.001	−0.015**	−0.023***	−0.009	−0.007	−0.020***	1.000		
FireCost	−0.030***	−0.034***	−0.005	−0.046***	−0.012**	−0.010*	−0.004	0.013**	1.000	
Flex	0.012**	0.013**	−0.007	0.016***	0.001	0.003	0.001	−0.402***	0.006	1.000

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Note: ***, ** and * denote significance at the 1%, 5% and 10% levels. Variance inflation factors computed after the two-way within transformation have a maximum of 1.30 and a mean of 1.04.

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4.3. Uncertainty and Entry-Level Hiring

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Table 4 reports the baseline. Column (1) includes only the uncertainty measure and the two sets of fixed effects: the coefficient is −1.193. Adding the full control vector in column (2) leaves it unchanged, which is the first indication that the estimate is not being driven by observable firm characteristics. Column (3) adds the level of generative-AI adoption. This is the decisive specification for the argument of the paper, because it puts capability and dispersion in the same equation: the level enters positively and significantly (0.300), while uncertainty enters negatively and is 4.1 times larger in magnitude. Column (4) adds firm-level general economic policy uncertainty, which is itself negative (−0.184) and consistent with everything known about uncertainty and factor demand [29,30], but which is 6.7 times smaller than the generative-AI term. The coefficient of interest is stable across all four columns.

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In the full specification, one standard deviation of generative-AI uncertainty lowers entry-level vacancy creation by 1.230 vacancies per thousand employees (standard error 0.026, $t = 47.0$), which is 15.6% of the sample mean. Column (5) estimates the multiplicative model of Equation (9) by Poisson pseudo-maximum likelihood on the vacancy count, with log lagged employment as an offset-like regressor: the semi-elasticity is −0.148, so a one-standard-deviation increase in uncertainty reduces entry-level postings by 13.8%. Evaluated at the sample mean rate that is a marginal effect of −1.164 vacancies per thousand employees, close to the least-squares estimate of −1.230; the remaining gap is what one expects when one estimator weights firms by their vacancy counts and the other does not. Column (6) turns to the stock: the share of employees aged 30 or below falls by 0.564 percentage points per standard deviation, 2.7% of its mean. Hypothesis 1 is supported, and the flow result shows up in the stock, as it must if the vacancy measure is capturing hiring rather than advertising.

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Table 4. Generative-AI uncertainty and entry-level vacancy creation.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
<i>AIU</i>	−1.193*** (0.027)	−1.193*** (0.026)	−1.251*** (0.026)	−1.230*** (0.026)	−0.148*** (0.004)	−0.564*** (0.018)
<i>AIE</i>			0.301*** (0.020)	0.300*** (0.020)	0.031*** (0.002)	0.088*** (0.018)
<i>EPU</i>				−0.184*** (0.020)	−0.022*** (0.002)	−0.120*** (0.019)
<i>Controls</i>	No	Yes	Yes	Yes	Yes	Yes
<i>Firm FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	29,660	29,660	29,660	29,660	29,660	29,660
<i>Within R²</i>	0.125	0.268	0.275	0.277	0.481	0.165

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Note: The dependent variable is the entry-level vacancy rate in columns (1) to (4), the entry-level vacancy count in column (5) and the share of employees aged 30 or below in column (6). Columns (1) to (4) and (6) estimate Equation (8) by least squares; column (5) estimates Equation (9) by Poisson pseudo-maximum likelihood with two absorbed effect dimensions and reports a semi-elasticity, and the statistic reported for it is a deviance-based pseudo R-squared. Standard errors clustered at the firm level are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels.

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4.4. Is the Chill Specific to the Young?

The single most informative test in the paper is the comparison of entry-level with experienced vacancies, because the two are posted by the same firms in the same years and are subject to the same demand shocks, the same financing conditions and the same managerial mood. What differs is the sunk component. Table 5 reports it. Column (1) repeats the baseline, -1.230 . Column (2) replaces the dependent variable with the experienced vacancy rate: the coefficient is -0.090 (0.016), 13.7 times smaller in magnitude. Columns (3) and (4) repeat the comparison with the Poisson estimator and find the same pattern, -0.148 against -0.008 .

The last rows of Table 5 report the cross-equation test of Equation (11). Stacking the two equations and clustering on the firm across both, the difference in the uncertainty coefficient is 1.140 with a standard error of 0.030, so equality is rejected at any conventional level. This is difficult to reconcile with any account in which generative-AI uncertainty is a proxy for general pessimism, poor performance or a common demand shock, since none of those is specific to the least experienced margin. It is exactly what Equation (6) predicts: the hurdle rises with dispersion in proportion to what the firm cannot get back, and what it cannot get back from a school leaver is much larger than what it cannot get back from a mid-career hire. Hypothesis 5 is supported.

Table 5. Is the chill specific to the young? Entry-level against experienced vacancies.

Variable	(1)	(2)	(3)	(4)
<i>AIU</i>	-1.230^{***} (0.026)	-0.090^{***} (0.016)	-0.148^{***} (0.004)	-0.008^{***} (0.001)
<i>AIE</i>	0.300^{***} (0.020)	0.275^{***} (0.017)	0.031^{***} (0.002)	0.034^{***} (0.001)
<i>EPU</i>	-0.184^{***} (0.020)	-0.120^{***} (0.018)	-0.022^{***} (0.002)	-0.013^{***} (0.001)
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Firm FE</i>	Yes	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes	Yes
<i>Observations</i>	29,660	29,660	29,660	29,660
<i>Within R²</i>	0.277	0.189	0.481	0.480
<i>Cross-equation difference</i>	1.140^{***} (0.030)			

Note: The dependent variable is the entry-level vacancy rate in column (1), the experienced vacancy rate in column (2), and the corresponding counts in columns (3) and (4), estimated by Poisson pseudo-maximum likelihood. The cross-equation difference is the coefficient on the interaction of *AIU* with the equation indicator in the stacked model of Equation (11), whose covariance is clustered on the firm across both equations. Standard errors clustered at the firm level are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels.

4.5. Irreversibility

Table 6 tests Hypothesis 2 by interacting uncertainty with each of the three measures of how sunk an entry-level hire is. Column (1) uses firm-specific training intensity. The interaction is -0.686 (0.021), so the chill is -0.549 for a firm one standard deviation below the mean of training intensity and -1.920 for a firm one standard deviation above it — a ratio of 3.5. Firms that invest most in turning a recruit into their own kind of employee are the firms that

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stop recruiting first, which is the uncomfortable core of the result: the training that makes entry-level employment valuable is also what makes it fragile.

Column (2) uses the provincial employment-protection index. The interaction is -0.300 (0.025), and the chill rises from -0.934 where enforcement is lenient to -1.534 where it is strict. Employment protection is not acting here as a tax on the level of employment, which is the classical reading [23], but as a multiplier on the effect of uncertainty: it does not make hiring dearer, it makes the hiring decision harder to undo, and it is reversal that the option is about.

Column (3) uses the reversible-employment margin. The interaction is 0.624 (0.023), of the opposite sign to the other two, and the chill falls from -1.856 for firms with little access to dispatched and outsourced labour to -0.609 for firms with much. A firm that can reverse the decision does not need to wait. Column (4) enters all three interactions together and each retains its sign and significance, so they are not proxies for one another. Hypothesis 2 is supported on all three margins.

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Table 6. Irreversibility: the chill and the sunk component of an entry-level hire.

Variable	(1)	(2)	(3)	(4)
<i>AIU</i>	-1.235^{***} (0.021)	-1.234^{***} (0.025)	-1.233^{***} (0.022)	-1.239^{***} (0.019)
<i>AIU × SpecTrain</i>	-0.686^{***} (0.021)			-0.514^{***} (0.019)
<i>AIU × FireCost</i>		-0.300^{***} (0.025)		-0.306^{***} (0.018)
<i>AIU × Flex</i>			0.624^{***} (0.023)	0.423^{***} (0.021)
<i>AIE</i>	0.290^{***} (0.019)	0.300^{***} (0.020)	0.295^{***} (0.019)	0.290^{***} (0.019)
<i>EPU</i>	-0.177^{***} (0.020)	-0.184^{***} (0.020)	-0.175^{***} (0.020)	-0.173^{***} (0.020)
<i>Effect at moderator = −1 SD</i>	-0.549^{***}	-0.934^{***}	-1.856^{***}	
<i>Effect at moderator = +1 SD</i>	-1.920^{***}	-1.534^{***}	-0.609^{***}	
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Firm FE</i>	Yes	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes	Yes
<i>Observations</i>	29,660	29,660	29,660	29,660
<i>Within R²</i>	0.319	0.286	0.312	0.340

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Note: The dependent variable is the entry-level vacancy rate and the model is Equation (10). The two rows of marginal effects report $\alpha_1 + \alpha_4 Z$ evaluated one standard deviation below and above the mean of the moderator used in that column. Standard errors clustered at the firm level are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels.

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4.6. The Contingent Turn

If uncertainty raises the hurdle for a committed hire by more than it raises the hurdle for a revocable one, firms should not only post fewer entry-level vacancies but post different ones. Table 7 tests this. Column (1) regresses the contingent share of entry-level vacancies on uncertainty with controls and both sets of effects, and column (2) adds the level of adoption and policy uncertainty: the coefficient is 2.245 (0.051), so one standard deviation of generative-AI uncertainty moves 2.25 percentage points of entry-level hiring

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from committed to revocable forms, 11.5% of the mean contingent share. Hypothesis 3 is supported.

The pattern of moderation confirms that this is the same mechanism seen from the other side. Column (3) shows that the compositional shift is larger for firms with an established reversible margin (1.250), which is where the substitution is cheapest to make, and column (4) shows that it is smaller for firms whose entry-level work is training intensive (−1.074) — precisely the firms for which a revocable contract is a poor substitute, because the value of the position depends on the recruit staying long enough to learn. Those firms have no second-best option, which is why column (5), which returns to the vacancy rate itself, finds that the chill on the total is attenuated precisely where the reversible margin is available (0.624 on the interaction). Taken together, columns (1) to (5) say that a firm under technological uncertainty does not withdraw from the youth labour market; it changes the terms on which it enters. This has a direct implication for measurement. A study that counts young employees will understate the change, because much of the adjustment is in the contract rather than in the headcount, and the welfare consequences of the two are not the same [41,43,58].

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Table 7. The contingent turn: the composition of entry-level hiring.

Variable	(1)	(2)	(3)	(4)	(5)
<i>AIU</i>	2.344*** (0.051)	2.245*** (0.051)	2.240*** (0.044)	2.238*** (0.046)	−1.233*** (0.022)
<i>AIU × Flex</i>			1.250*** (0.042)		0.624*** (0.023)
<i>AIU × SpecTrain</i>				−1.074*** (0.044)	
<i>AIE</i>		0.352*** (0.045)	0.343*** (0.045)	0.337*** (0.045)	0.295*** (0.019)
<i>EPU</i>		0.263*** (0.046)	0.281*** (0.046)	0.274*** (0.046)	−0.175*** (0.020)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes
<i>Firm FE</i>	Yes	Yes	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	29,660	29,660	29,660	29,660	29,660
<i>Within R²</i>	0.122	0.126	0.157	0.149	0.312

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Note: The dependent variable is the contingent share of entry-level vacancies in columns (1) to (4) and the entry-level vacancy rate in column (5). Standard errors clustered at the firm level are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels.

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4.7. Inaction and Hysteresis

Table 8 tests the band. Column (1) regresses the inaction indicator on uncertainty with controls, and column (2) adds the level of adoption and policy uncertainty: one standard deviation raises the probability that the firm makes no adjustment at all to its young workforce by 0.050 (0.003), which is 28.5% of the unconditional inaction rate of 17.7%. Neither the level of adoption nor policy uncertainty has any comparable effect, which is what distinguishes a widening band from a shift in the desired level: a shift in the desired level moves firms out of the band on one side, whereas a wider band traps them inside it whatever their desired adjustment. Column (3) shows that the band widens most for firms with training-intensive entry-level work (0.029, p =

0.000), as Equation (6) implies, since the two triggers that bound the band both move with the sunk cost.

Column (4) restricts attention to firms that were inactive in the previous year — 4,707 firm-years, of which 41.3% resume expansion — and asks what makes them move. Uncertainty lowers the probability of resuming by 0.126 (0.014). This is hysteresis in the precise sense of the real-options literature: the firm that has stopped adjusting requires a larger shock to start again than it required to stop [20,22]. Column (5) reports the underlying net change itself, which falls by 0.439 percentage points of lagged employment. Hypothesis 4 is supported. The finding has an uncomfortable corollary for policy: a measure that reduces uncertainty after firms have entered the band has to overcome the band's width, not merely restore the previous level of confidence.

Table 8. The zone of inaction and its hysteresis.

Variable	(1)	(2)	(3)	(4)	(5)
<i>AIU</i>	0.050*** (0.003)	0.050*** (0.003)	0.051*** (0.003)	−0.126*** (0.014)	−0.439*** (0.011)
<i>AIU × SpecTrain</i>			0.029*** (0.003)		
<i>AIE</i>		−0.005* (0.003)	−0.004 (0.003)	−0.001 (0.015)	0.060*** (0.011)
<i>EPU</i>		0.001 (0.003)	0.001 (0.003)	−0.015 (0.016)	−0.062*** (0.012)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes
<i>Firm FE</i>	Yes	Yes	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	29,660	29,660	29,660	4,707	29,660
<i>Within R²</i>	0.016	0.016	0.022	0.085	0.092

Note: The dependent variable is an indicator for no adjustment to the young workforce in columns (1) to (3), an indicator for a positive net change in column (4), estimated on the subsample of firms that were inactive in the previous year, and the net change itself in column (5). Columns (1) to (4) are linear probability models with firm and year effects absorbed. Standard errors clustered at the firm level are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels.

4.8. Endogeneity

The estimate so far could be contaminated by a firm-year shock that raises both the willingness to hire and the amount managers write about the technology — a firm that is expanding into a new product line will do both. Such a shock biases the coefficient towards zero, so the least-squares estimate is a lower bound in absolute value. Table 9 reports the instrumental-variables estimates. Panel A gives the first stage. The shift-share instrument built from pre-sample occupational exposure enters with a coefficient of 0.444, the frontier-release instrument with 0.082 and the provincial peer instrument with 0.421, all significant at the 1% level, and the Kleibergen–Paap rank Wald statistic is 5353.0, far above any conventional critical value for weak instruments [66].

Panel B gives the second stage. Column (1) repeats the least-squares estimate for comparison, −1.230. Column (2) is two-stage least squares: −1.302 (0.037), larger in magnitude than the least-squares estimate by 0.072, in the direction the attenuation argument predicts. The Hansen test of the overidentifying restrictions does not reject ($p = 0.091$), so the three

instruments agree on the parameter they are estimating, and the Wu-Hausman test rejects exogeneity ($p = 0.0007$), which is why the two estimates differ. Column (3) reports the two-step generalised method of moments estimator, -1.303 , essentially identical. Columns (4) and (5) instrument the same regressor in the contingent-share and experienced-vacancy equations: the contingent turn survives instrumentation (2.211) and the experienced-vacancy coefficient remains an order of magnitude smaller (-0.104). We treat the instrumented estimates as the preferred ones and the least-squares estimates as conservative.

Table 9. Endogeneity: instrumental-variables estimates.

Variable	(1)	(2)	(3)	(4)	(5)
<i>Panel A: first stage, dependent variable AIU</i>					
IV^{Shift}	0.444*** (0.032)				
IV^{Front}	0.082*** (0.031)				
IV^{Peer}	0.421*** (0.004)				
Kleibergen-Paap rk Wald F	5353.0				
<i>Panel B: second stage</i>					
AIU	-1.230*** (0.026)	-1.302*** (0.037)	-1.303*** (0.037)	2.211*** (0.075)	-0.104*** (0.025)
AIE	0.300*** (0.020)	0.314*** (0.020)	0.315*** (0.020)	0.359*** (0.047)	0.278*** (0.018)
EPU	-0.184*** (0.020)	-0.175*** (0.021)	-0.174*** (0.021)	0.268*** (0.047)	-0.119*** (0.018)
Estimator	OLS	2SLS	GMM	2SLS	2SLS
Hansen J (p)		0.091	0.091	0.301	0.582
Wu-Hausman (p)		0.0007		0.4732	0.4436
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	29,660	29,660	29,660	29,660	29,660

Note: Panel A reports the first stage of Equation (12); Panel B reports Equation (13). The dependent variable is the entry-level vacancy rate in columns (1) to (3), the contingent share in column (4) and the experienced vacancy rate in column (5). The Hansen statistic tests the overidentifying restrictions and the Wu-Hausman statistic tests the exogeneity of AIU. Standard errors clustered at the firm level are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels.

4.9. The Frontier Releases as an Event

The identification so far uses cross-firm variation in uncertainty within a year. A complementary design uses the timing of the technology itself. Table 10 estimates Equation (14), comparing firms in the top tertile of pre-sample occupational exposure with the rest, year by year, with 2021 as the reference. The leads are the test that matters. None of the pre-shock interactions is significant and the joint test of the 5 leads does not reject (chi-squared = 3.99, $p = 0.551$), so highly exposed and less exposed firms were not on different entry-level hiring trajectories before the technology arrived.

653 The lags are large and immediate. Relative to 2021, entry-level vacancy
654 creation in exposed firms falls by 1.65 vacancies per thousand employees in
655 2023 (0.169), by 2.14 in 2024 and by 1.97 in 2025. The 2022 coefficient is
656 already negative but much smaller, which is what one would expect of a
657 reporting year that ended weeks after the first public release. Column (2)
658 repeats the design for the contingent share and finds the mirror image: leads
659 that are jointly indistinguishable from zero ($p = 0.439$) and a rise of 3.83
660 percentage points by 2024. The two designs — cross-firm variation in
661 disclosed uncertainty and time-series variation in the arrival of the frontier —
662 point to the same conclusion from independent sources of variation.

663 **Table 10.** Dynamic difference in differences around the frontier-model releases.

Variable	(1) Entry rate	(2) Contingent share
<i>Treat x 2016</i>	−0.006 (0.183)	0.258 (0.396)
<i>Treat x 2017</i>	0.073 (0.169)	0.361 (0.385)
<i>Treat x 2018</i>	0.267 (0.183)	0.148 (0.374)
<i>Treat x 2019</i>	−0.064 (0.171)	0.784** (0.376)
<i>Treat x 2020</i>	0.078 (0.163)	0.351 (0.371)
<i>Treat x 2022</i>	−0.309* (0.162)	0.426 (0.363)
<i>Treat x 2023</i>	−1.654*** (0.169)	2.998*** (0.371)
<i>Treat x 2024</i>	−2.136*** (0.167)	3.834*** (0.373)
<i>Treat x 2025</i>	−1.971*** (0.167)	4.164*** (0.385)
<i>Joint test of pre-shock leads, χ^2</i>	3.99	4.82
<i>Joint test of pre-shock leads, p</i>	0.551	0.439
<i>Controls</i>	Yes	Yes
<i>Firm FE</i>	Yes	Yes
<i>Year FE</i>	Yes	Yes
<i>Observations</i>	29,660	29,660
<i>Within R^2</i>	0.169	0.042

664 Note: The model is Equation (14). Treated firms are those in the top tertile of pre-sample
665 occupational exposure to language models and 2021 is the reference year. The joint test of pre-
666 shock leads is a Wald test that the interactions for 2016 to 2020 are jointly zero. Standard errors
667 clustered at the firm level are in parentheses. ***, ** and * denote significance at the 1%, 5% and
668 10% levels.

669 5. Further Investigations

670 5.1. Robustness

671 Table 11 reports 12 alternative specifications. Replacing the year effects with
672 industry-by-year and then province-by-year effects, which absorb any shock
673 common to an industry or a province in a given year, leaves the coefficient
674 essentially unchanged. Clustering on both the firm and the year rather than on
675 the firm alone widens the standard error but leaves the estimate significant at
676 the 1% level. Estimating in logs of the count, restricting to the balanced panel,

dropping the pandemic years 2020 and 2021, dropping the final year, dropping state-owned firms, winsorising the dependent variable at the 5th and 95th percentiles, restricting to firms with at least five annual reports, and adding the lagged dependent variable all leave the conclusion intact. Across the specifications measured in the units of the baseline the coefficient ranges from -1.239 to -1.104 .

Three further checks address inference and omitted variables rather than specification. First, although the panel has 3,200 clusters, we recompute the p-value of the coefficient by a restricted wild cluster bootstrap with Rademacher weights and 999 replications [72,73]: $p = 0.001$. Second, we permute the uncertainty measure at random within each year 300 times and re-estimate; the actual coefficient is more extreme than the permuted ones with a randomisation p-value of 0.003. Third, we apply the coefficient-stability bound of Oster [74]. Moving from the specification without controls to the specification with them changes the coefficient from -1.228 to -1.230 while the R-squared rises from 0.135 to 0.277; the coefficient is therefore almost completely insensitive to the observed controls, the bias-adjusted value is -1.231 , and selection on unobservables would have to be 1368 times as important as selection on the observed controls, and to work in the opposite direction, to drive the estimate to zero.

One timing check is worth reporting separately because it is a test of the mechanism rather than of robustness. Entering the contemporaneous and the once-lagged uncertainty measure together, the contemporaneous term retains a coefficient of -1.220 while the lagged term is -0.007 (0.020). It is current dispersion that sets the current hurdle, exactly as Equation (4) states: the trigger is a function of today's σ , not of last year's, and a firm that has become confident again should hire again. That the effect is contemporaneous is also why the policy lever is fast-acting.

Table 11. Robustness checks.

Specification	AIU	SE	N	Within R ²
Baseline	-1.230^{***}	(0.026)	29,660	0.277
Industry x year effects	-1.217^{***}	(0.027)	29,660	0.267
Province x year effects	-1.224^{***}	(0.028)	29,660	0.256
Two-way clustering (firm and year)	-1.230^{***}	(0.035)	29,660	0.277
Dependent variable in logs	-0.180^{***}	(0.004)	29,660	0.308
Balanced panel	-1.204^{***}	(0.029)	24,240	0.274
Excluding the pandemic years	-1.239^{***}	(0.029)	23,390	0.280
Excluding 2025	-1.230^{***}	(0.028)	26,621	0.275
Excluding state-owned firms	-1.124^{***}	(0.030)	20,484	0.254
Winsorised at the 5th and 95th percentiles	-1.104^{***}	(0.022)	29,660	0.300
Firms with at least five annual reports	-1.228^{***}	(0.026)	29,621	0.277
Adding the lagged dependent variable	-1.222^{***}	(0.028)	26,460	0.280

Note: Each row is a separate estimation of Equation (8) with the modification named in the first column; the coefficient reported is that on AIU. The specification in logs uses the logarithm of one plus the entry-level vacancy count and is therefore not in the units of the other rows. Standard errors clustered at the firm level are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels.

5.2. Heterogeneity

Table 12 splits the sample four ways. The chill is stronger in state-owned firms (-1.467) than in private ones (-1.124), with a difference of -0.343 . This is consistent with the irreversibility reading rather than with a story about

financial constraints, since state-owned firms in China face softer budget constraints but much harder employment ones: the position they open is the position they keep.

The chill is stronger in technology-intensive firms (−1.567 against −1.085), where the entry-level task bundle is most directly in the path of the frontier and where the firm knows enough about the technology to be uncertain about it in an informed way. It is stronger where employment protection is strictly enforced (−1.483 against −0.974), which reproduces the interaction result of Table 6 in a sample split. And it is statistically indistinguishable across slack and tight local youth labour markets (difference −0.060, $p = 0.268$), which is informative: the mechanism is internal to the firm's investment problem and does not depend on how easy young workers are to find. A demand-side story in which firms cut junior hiring because juniors are cheap to replace later would predict the opposite.

Table 12. Heterogeneity analysis.

Partition	Subsample A	AIU	N	Subsample B	AIU	N	Difference
State ownership	State-owned	−1.467*** (0.050)	9,176	Non-state	−1.124*** (0.030)	20,484	−0.343*** (0.058)
Technology intensity	High technology	−1.567*** (0.048)	8,879	Other	−1.085*** (0.031)	20,781	−0.482*** (0.057)
Employment protection	Strict	−1.483*** (0.036)	14,834	Lenient	−0.974*** (0.036)	14,826	−0.509*** (0.051)
Local youth labour market	Slack	−1.247*** (0.037)	14,830	Tight	−1.187*** (0.039)	14,830	−0.060 (0.054)

Note: Each partition re-estimates Equation (8) on the two subsamples. The difference is the coefficient in subsample A minus that in subsample B, with a standard error computed from the two independent estimates. Standard errors clustered at the firm level are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels.

5.3. Economic Magnitudes

It is worth stating the size of the effect in units a firm or a ministry would recognise. The median firm in the sample employs 2,089 people, so one standard deviation of generative-AI uncertainty removes about 2.6 entry-level vacancies a year for such a firm at the least-squares estimate and 2.7 at the instrumented one. Moving a firm from the first to the third quartile of the uncertainty distribution, a gap of 1.30 standard deviations, costs 1.60 vacancies per thousand employees against a mean rate of 7.87, or 20% of entry-level vacancy creation, and raises the contingent share by 2.92 percentage points.

The episode itself can be sized from the event study rather than from the cross-section, which is the cleaner calculation because the mean of a standardised measure cannot move. What the arrival of the technology did was to pull firms apart: the cross-sectional standard deviation of the uncertainty measure rose from 0.86 before 2023 to 1.25 after, and the gap in the measure between highly exposed and less exposed firms widened by 1.56 standard deviations. Table 10 prices that widening directly: exposed firms lost 2.14 entry-level vacancies per thousand employees by 2024 relative to 2021 and to less exposed firms, which is 27% of the sample mean rate, and gained 3.83 percentage points of contingent share over the same interval. The observed unconditional entry-level vacancy rate fell from 7.98 to 7.61 across

the break, so the exposure-driven component accounts for a substantial part of an aggregate movement that would otherwise be attributed to demand. The quantities are large enough to matter for a cohort and small enough to be plausible, and they are of the same order as the relative employment declines documented for exposed young workers in the United States [7].

6. Discussion

The central claim of this paper is that the entry-level labour market responds to generative artificial intelligence before the technology does anything to it. The channel is the option value of waiting, and the evidence for it is not the sign of a single coefficient but the pattern of five: the effect is driven by dispersion rather than by level; it is concentrated on the margin with the largest sunk component; it scales with measures of irreversibility and is attenuated by measures of reversibility; it produces a compositional shift towards revocable contracts rather than only a reduction in quantity; and it generates a zone of inaction with hysteresis. No one of these is decisive on its own. Together they are difficult to produce with any account that does not have an option in it.

This changes how the existing evidence should be read. The literature has been arguing about whether GenAI destroys entry-level jobs, with the implicit understanding that a positive finding would mean the technology is already doing the work. Our results suggest that a large part of the measured decline can occur while the technology is doing nothing at all, purely because firms do not yet know what it will do. That is not a reason to be sanguine — a cohort that enters the labour market during the wait bears the same scars as a cohort that enters during a recession [37–39] — but it changes the diagnosis and therefore the treatment. It also implies that estimates of the employment effect of AI capability that use the post-2022 period are contaminated by the option-value response, and that the two need to be separated before either is interpreted structurally.

For firms, the results identify a decision that is being made by default. The option-value logic is individually rational: a firm that does not know what its junior analysts will be doing in three years is right to hesitate before sinking three years of training into them. But the hesitation has a cost the firm does not see on its own books. An entry port that stays closed for three years leaves the firm, at the end of them, with no cohort in its middle ranks, and the internal labour market cannot be restarted on demand [75]. Recent systems-level work on Chinese firms points the same way, finding that the returns to artificial intelligence depend on the organisational and innovation capabilities that surround it [76–79]. The instruments that restore reversibility — modular induction, portable rather than firm-specific early training, rotation across task bundles — are also the instruments that lower k and therefore the wedge in Equation (4). Our moderation results say that firms which already have them behave very differently from firms which do not.

For policy, the target follows directly from the model. Almost everything currently proposed under the heading of AI and employment addresses capability: retraining for the tasks the technology will take, subsidies for the tasks it will create. The variable in Equation (5) is not capability but its dispersion, and dispersion is reducible by public action in a way that capability is not. Sectoral technology roadmaps, disclosure standards for model capabilities and limitations, and certification of what models can and cannot

be relied upon in regulated work all narrow σ without touching μ . The responsible-governance literature has argued for such instruments on grounds of accountability [80,81]; the present results suggest they are also youth employment policy, and unusually cheap ones. The contingent-turn result adds a second and more uncomfortable target. If firms respond to uncertainty by moving young workers onto revocable contracts, then a policy that measures success by youth headcount will record success while the quality of entry employment deteriorates. Whether that matters depends on whether such contracts are stepping stones or dead ends, which is exactly the question the contingent-work literature has not settled [41–43,58].

Three limitations bound the claims. First, disclosure-based uncertainty measures what management writes, and firms differ in how much they write; the firm fixed effects absorb the persistent part of that, and the instruments address the transitory part, but the measure is a proxy. Second, the sample is listed firms in one country, which are larger, more visible and more exposed to disclosure incentives than the firms that employ most young workers; the direction of the bias is not obvious, since unlisted firms face weaker employment protection but also have less capacity to bear the option. Third, the design identifies the effect of uncertainty over a four-year window after the shock, and cannot say what happens when the frontier settles — whether the deferred hiring is made up, which the option-value model predicts, or whether the entry port has been re-engineered in the interval, which the organisational literature on technology adoption would suggest [82–86]. Following the same firms as σ falls is the obvious next study, and the fall, when it comes, will be the cleanest test of the mechanism available.

7. Conclusions

This paper asked why entry-level employment should fall in response to a technology that experiments show to be most useful to the least experienced. The answer offered here is that firms are not responding to what the technology does but to what they do not know about what it will do. Hiring a young worker is an irreversible investment in a task bundle whose future is uncertain, and the theory of investment under uncertainty says that the firm should then wait, that it should wait longer the more of the investment it cannot recover, and that it should prefer whatever form of the investment it can undo.

Using 29,660 firm-year observations for 3,200 Chinese A-share listed firms over 2016–2025, we find all three. One standard deviation of firm-level generative-AI uncertainty lowers entry-level vacancy creation by 1.230 vacancies per thousand employees, 15.6% of the mean, while the level of adoption enters with the opposite sign and general policy uncertainty is 6.7 times smaller; the effect is 13.7 times larger for entry-level than for experienced vacancies; it scales with firm-specific training intensity and with employment protection and is attenuated by access to reversible employment; it raises the contingent share of entry-level vacancies by 2.25 percentage points; and it widens a zone of inaction, raising the probability of no adjustment by 0.050 and lowering the probability of resuming expansion by 0.126. Instrumental variables, a dynamic difference-in-differences design with flat pre-shock leads, a wild cluster bootstrap, randomisation inference and coefficient-stability bounds all support the reading.

The policy conclusion is unusually specific. If the entry port is closing because firms are uncertain rather than because they have been displaced, then the cheapest available intervention is not retraining but information: anything that tells firms what the technology will and will not do in their sector narrows the dispersion that is doing the damage. And the measurement conclusion is equally specific. Counting young employees will miss most of what is happening, because the first thing a firm gives up under uncertainty is not the young worker but the commitment to them.

Appendix A

Table A1 reports placebo and alternative-measure checks. The first row repeats the baseline. The next rows enter general economic policy uncertainty and the adoption level on their own, so that their unconditional associations can be compared with the conditional ones in Table 4; the experienced-vacancy result of Table 5 is repeated for convenience. Total employment growth shows no significant response, which is the sharpest statement of the compositional nature of the finding: the firm is not shrinking, it is changing whom it hires. Capital expenditure responds negatively but by a small amount, which is not a placebo failure but the classical uncertainty-and-investment result [21,29] showing up where it should, and its smallness relative to the entry-level effect is itself informative about which irreversible investment is most exposed. The last three rows replace the disclosure measure with the dispersion of analyst forecasts for generative-AI-exposed segments, with a count of risk-factor sentences mentioning the technology, and with an indicator for the top quartile of the baseline measure; all three reproduce the result.

Table A1. Placebo tests and alternative measures.

Specification	Coefficient	SE	N	Within R ²
Entry-level vacancy rate on AI uncertainty (baseline)	−1.230***	(0.026)	29,660	0.277
Placebo: general economic policy uncertainty only	−0.344***	(0.022)	29,660	0.151
Placebo: AI adoption level only	0.041*	(0.022)	29,660	0.143
Placebo: experienced vacancy rate	−0.090***	(0.016)	29,660	0.189
Placebo: total employment growth	0.093	(0.118)	26,460	0.106
Comparison: capital expenditure rate	−0.041***	(0.016)	29,660	0.010
Uncertainty measured as the dispersion of analyst forecasts	−1.040***	(0.024)	29,660	0.245
Uncertainty measured by a count of risk-factor sentences	−1.094***	(0.025)	29,660	0.254
Uncertainty entered as an indicator for the top quartile	−1.922***	(0.050)	29,660	0.213

Note: The coefficient reported is that on the variable named in the first column, or on AIU where no other variable is named. All specifications include the full control vector and firm and year effects. Standard errors clustered at the firm level are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels.

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